

Matthew Barnes

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EDUCATION

Washington State University, BS Computer Science
Major GPA: 3.3

Expected Graduation: Dec 2026

PROJECTS

Web Development | **Flick Finder – Movie Suggestion Website (JavaScript, HTML, CSS)** [GitHub](#)

- Utilizes the OpenAI API to generate personalized movie recommendations based on user input.
- OpenAI API model: gpt-3.5-turbo. Designed for contextual understanding, allowing for accurate suggestions.
- Integrated TMDB API to fetch and display appropriate movie posters aligned with OpenAI API's suggestions.
- Designed a responsive UI using modern CSS.

Collaborative Software Development | **Battleship (C++, SFML)** [GitHub](#)

- Led the development of the game board, menu, and ship mechanics, ensuring seamless user experience.
- Implemented branching strategies and utilized Git & GitHub for version control, maintaining code integrity and efficient team collaboration.
- Coordinated with team members through code reviews, enhancing code quality and maintainability.

Data Structures & File Parsing | **Amazon Inventory REPL (C++)** [GitHub](#)

- Developed a REPL-based inventory system using custom hash tables and linked lists for efficient product indexing and lookup from CSV data.
- Enabled real-time product search, insertion, and inventory management through a command-line interface.
- Parsed and managed 1000+ product records

Mobile Development | **Random Location Guessing Game (Swift, SwiftUI, Google Maps SDK)** [GitHub](#)

- Built an interactive geography game in SwiftUI leveraging Google Maps SDK for embedded Street View navigation and location-based challenges.
- Developed a reusable game engine supporting multiple modes with randomized location logic and map-based gameplay.
- Applied CoreLocation and Google APIs to present visual clues and calculate distances between guesses and actual coordinates.

SKILLS

Languages: C, C++, C#, Python, Swift, JavaScript, HTML, CSS

Tools: Git, GitHub, Visual Studio, MS Office

Platforms: macOS, Linux/Unix, Windows

RELEVANT COURSE WORK

Computer Programming | Program Design & Development | Data Structures & Algorithms | Advanced Data Structures | Intro to Computer Architecture | Linear Algebra | Calculus I | Calculus II | Calculus III

In progress (Fall 2025): Automata & Formal Languages | Software Engineering I | Intro to Machine Learning